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Older adults consistently reject digital technology even when designed to be accessible and trustworthy.

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The Wisdom of Older Technology (Non)Users

IT IS IMPOSSIBLE not to notice that many of the questions driving research on technology use by older adults today are the same as those at the forefront of aging and accessibility research 20 years ago. Back then, computers were predominantly large desktops, social media was still on the horizon, and mobile phones were large and not (yet) smart. Older adults had little presence on the Internet. Today, devices have changed and older adults are increasingly online.^{9,15} They do, however, continue to lag in broadband use, breadth of applications used, and time online.¹² Typical reports reflect they have little interest in social media (other than staying in touch with family) and are skeptical of online financial transactions.¹⁷

Clearly, the problem of older adults' comparatively limited technology use has not gone away despite

a more tech-savvy group of people aging into the "older adult" category. According to the most recent data, from 2014–2016,^{12,15} predictions of a forthcoming "Silver Tsunami" of retired workers—a cohort now accustomed to digital technology access in their working lives and therefore able to take full advantage of the Internet—have not come true.⁹ Indeed, the overwhelming perception remains of older generations being incapable of or otherwise resistant to using technology. A "digital divide" between old and young is potentially more disabling now compared to 20 years ago, given the push for a more fully realized digital society. Digital technologies today are so essential to daily life that it is reasonable to ask whether older adults' inability to access online-only government services may soon be included among the precipitating factors in older adults moving into assisted living.

While we see the emergence of calls for a more holistic view of how to design technology for older adults^{4,13,18} than was the case 20 years ago, interventions to get older adults online commonly focus on age-related declines (such as vision, hearing, cognition, and dexterity) as the principal barriers to technology adoption. These interventions are often senior-friendly variants or adaptations to make the

» key insights

- **Older adults' non-use of digital technologies is purposeful and thus instructive for identifying problematic consequences of these technologies for the population at large.**
- **Looking beyond traditional thinking on how to make it easier for older adults to use technology, we identify factors relating to responsibility, values, and cultural expectations contributing to older adults' resistance to digital technologies.**
- **These factors emerge from the cultural changes driven by technological innovation, so will likely remain barriers to adoption, even as younger generations age, unless new technologies are designed with sensitivity to values fostered through such experience.**



technology more accessible.^{6,10} However, older adults are considerably less likely than their younger counterparts with a disability to adopt assistive tools designed specifically for them,³ suggesting perhaps they do not view the conditions of aging as disabling,¹⁰ or (or in addition) their resistance to technology adoption is not solely or even primarily rooted in usability/accessibility issues¹⁹ (see the sidebar “Who Is an ‘Older Adult?’”).

Our interviews with older adults reveal they are often unwilling to acknowledge that their lives would be enriched through digital technologies, whether or not they were made accessible. It is this attitude concerning technology that intrigues us. Given that the kinds of technologies and applications older adults are receptive to or averse to varies by individual, older users do not appear to be identifying inherent de-

sign failings of any specific tools. Are there bigger-picture issues with “digital society” that lead older adults to reject particular technologies? If so, are they likely to be of continuing relevance when future generations age into older adults? And is there anything that can be done to address them?

Here, we draw from our own recent research interviews and a substantial body of experience working with older adults to describe three factors—responsibility, values, and cultural expectations—that contribute to older adults’ resistance to the digital proficiency that is ostensibly required to be fully participating, independent citizens in our increasingly digital society. These factors suggest new directions for aging and accessibility research, while also being more broadly instructive for creating a digital society that works for everyone.

Older Users’ Experience

While there is reason for optimism concerning older adults’ adoption of technology when looking at their increasing online participation,¹⁵ that participation is qualitatively different from younger users, being more limited in time and variety of experiences.¹² We sought to better understand the underlying reasons for these differences in a series of group interviews with a total of 14 post-retirement community-dwelling individuals, ages 66 to 86, around Dundee, Scotland, who we drew from an established older-adult participant pool.⁸ While these discussions revealed some physical and cognitive decline among participants, we were not aware of any having physical or cognitive deterioration outside the typical range for their age. The focus groups followed a semi-structured format that allowed significant con-

Who Is an ‘Older Adult’?

Various age groupings have been used over the years to define “older.” The fact is, aging is a process. Governments define age for pensions and Social Security, and various services offer senior rates based on age. From an individual’s point of view, however, age is largely a state of mind. A person who is 60 may feel old, while another who is 80 may not. Research on the use of technology by older adults has varied in terms of the cut points for age categorization. Over age 50 is often used, though a less-controversial cutoff would be age 65. As noted by one of our participants, people do not suddenly wake up one day and find themselves “old,” nor do they wake up to find they are no longer able to use technology. Age itself is not the sole criterion determining technology usage behavior, as there is great variability in adoption and use by those conventionally categorized as “older adults.” So while some broad assertions can be made about older adults in comparison with, say, younger adults, it is always important for researchers to be aware of the ways older adults are individuals.

versational steer by participants. Our conversations focused on participants’ use of the Internet, what they did not use it for, and what aspects of digital technologies they did or did not trust.

Overall, participants were open to using at least a limited set of applications. Email and general Web browsing were used by all. Social networking, travel booking, online shopping, and online banking were used by some, often to the point of dependence. Notably, participants did not consider learning and using technology rewarding in and of itself. Many also talked about consciously avoiding “getting caught up in” digital life, viewing the abundance of applications and features as potential diversions from more rewarding activities. Social networking often fell into the time-wasting category, with many noting the insipidness of the content on Facebook, though some found it useful (and even enjoyable) for keeping in contact with family. For those in the former camp, there was a strong aversion both to the idea of one’s life being an “open book” and being glued to one’s mobile phone—trends they found deeply troubling in younger generations.

Besides the limited range of tools the participants adopted, the most striking characteristic of their reported use was lingering discomfort. “Although I use the computer, I find it quite frightening,” admitted one woman. “The reason I find it frightening is that I don’t understand it. And I don’t know how to put things right.” They described feeling much more competent, and therefore more comfortable, with analog equivalents (such as paper ar-

chives and paper calendars). They described worrying about and planning for the eventuality of their computer “blowing up.” Security concerns were omnipresent. Even tools used regularly were not trusted per se. Rather, when they acknowledged significant benefits of specific tools, they used them in spite of unresolved concerns regarding their trustworthiness.

While none of what we found may be surprising, it is worth emphasizing that this pattern of use paints a picture that clashes with the dominant cultural narrative of older adults being resistant to all digital technologies by default. It also provides a more nuanced view of recent claims that uptake of digital technologies is rising among older adults; while a much greater percentage of older adults is online than a decade ago, they are very discriminating in what they are willing to do (see the sidebar “Why Focus on Nonuse?”). And, as we intend to show, much of what underlies the resistance is inherent in aspects of being older that are unlikely to change as new generations reach retirement age.

Underlying Problems

Here, we explore what underlies older adults’ resistance to the many digital tools that would ostensibly provide so many benefits to them (such as easing loneliness and isolation, being in control of decisions that affect them, living independently, and participating in and contributing to society).¹ We identify three clusters of factors that can contribute to resistance, though note their relevance and how their interactions play out differently within each individual.

Perception of risk. Upon retiring, people in the industrialized West lose an important training ground (and motivation) for developing competence with emerging technologies. One approach to addressing it is to create IT drop-in centers or training courses tailored to older users. Such resources are valuable for older adults seeking information relating to precise steps for executing a task, or procedural knowledge, as they so often ask for, as explored by Leung et al.,¹¹ but typically do not strengthen their conceptual grounding in ways that enable them to execute unfamiliar tasks. As a result, existing training opportunities for older adults do little to affect generalized anxiety about not “understanding” technologies. Most of our participants seemed to worry they did not know enough to use the tools effectively and responsibly and did not know how they would know when they did know enough.

A contributor to these feelings of incompetence was that in the past our participants would seek out trained professionals to accomplish specialty tasks for them; for example, they would go to a mortgage advisor to get advice on choosing a mortgage, to a travel agent to arrange hotels and flights, or to a banker to handle the transfer of money. A consequence of having more immediate “control”¹ over these tasks is having to take on new responsibilities, which some felt equated to “having a part-time job,” requiring hours in front of a computer screen. Though this may surprise many adults in full-time employment, older adults’ lives are still extremely busy, with clubs, activities, commitments to family and friends, and more mundane chores and responsibilities (such as home repair, medical appointments, and shopping). They simply do not have time to learn how to use online services well enough to use them with confidence.

In terms of online banking, a common response is, “I don’t trust it,” as in Vines et al.¹⁷ But upon further probing, it becomes clear that these older adults do not trust themselves. They lack confidence in their ability to use the tools and fear the consequences of making mistakes. What happens to their money if they press the wrong button? If they are hacked, will they be held ac-

countable for not following security protocols they ought to have known? They are right to worry in both cases. It is unclear what mistakes might be correctable and quite likely that more personal responsibility will be assumed as expectations for digital proficiency rise. We can hardly fault older adults for deciding it unwise to use any tool—online banking and shopping or submitting official government forms—without learning to use it in ways that ensure their safety and security.

The assurance of a clearly understandable safety net when conducting digital activities is essential for older adults to adopt tools, more so (with online financial transactions) because recovery from being defrauded of their financial savings would be much more difficult. In addition to developing a legal scaffolding for such a safety net or policies forcing businesses to assume the costs of user error, including accidental breaches of security protocols, there is important work to be done in devising mechanisms and user interactions that make data systems more (if never entirely) foolproof. Acknowledging that individuals often adopt a tool despite not trusting it, it is critical to design mechanisms that help manage user anxieties (such as providing necessary feedback and reassurances throughout the interactions) to ensure effective use and prevent panic and abandonment.

The value proposition. Not often discussed in the literature on aging and accessibility is that choosing not to use new technology can be seen as a rational decision for older adults, depending on their resources and needs; for example, among those who live on limited pensions, it may be difficult to justify the financial outlay for broadband alone. And in the case of a service like online shopping, while it would seemingly provide numerous benefits—saving time and money or not having to travel—it would also replace an important social activity for those who shop (sometimes daily) purely for the social benefit. Indeed, older adults we interviewed work hard to strike a positive balance between online proficiency and cultivation of rich offline social worlds.

A surprisingly common feature of our conversations with older adults

about reasons for resisting certain technologies was their strong sense of social responsibility. They worry, for example, that online shopping takes business away from local shops, meaning there will soon be no vibrant town centers in which to socialize with friends. They worry in particular that if they do not make an effort to attend, say, physical shops and banks, the people who work there will soon be out of their jobs. One participant expressed sympathy for the “delightful” receptionist at the sports facility whose job was replaced by an app for booking classes. Another said she would never pay her road tax or anything else online, not because of concern about using the technology but, “I just think I want to keep the post office open.”

If and when some of the digital technologies older adults resist become essential to daily (independent) living, it will be important to explore that resistance to understand what might motivate or enable them to use the technologies in the future. While older adults’ mental trade-offs—weighing the perceived benefits against the financial cost of the technologies, the time it will take to learn to use them, the social interactions they may lose, and the jobs the technologies replace—will be evaluated differently by different people, there are at least three things that must change to tip the scales. Broadband cannot continue to be charged at rates that are prohibitive for many older adults; this needs to be treated by government regulators and access providers as a basic need like electricity.² Moreover, with loneliness being such a common characteristic of the older-adult experience, greater attention needs to be paid to ensuring digital engagements

do not replace social interactions but instead where possible facilitate new social and community-building opportunities. Today, social networking systems (such as Facebook) fail to broadly address this need for older adults. Part of getting older adults online will also be developing strategies for creating new, good-quality jobs in place of those the digital technologies make redundant—something that, regardless of older adults’ attitudes toward technology, needs attention.¹⁶

Freedom of low expectations. The notion that aging per se leads to technology abandonment does not withstand scrutiny. And yet older adults themselves are often the worst perpetrators of the myth, quick to excuse their disinterest in a given tool with the seemingly self-explanatory line, “I’m too old.” It is worth considering, then, what older adults might gain from this stereotype.

We (the authors) have come to understand that it affords older adults the privilege of taking quiet personal stands against the aspects of technology they find worrying, threatening, or plain annoying. For example, a common justification for not using Facebook and other social media is the cyber-bullying, “stalking” behavior, and “narcissism” they seem to encourage. One older adult we talked to said, “I don’t do Facebook. Having been a teacher, I think it’s got loads of problems for young people. [T]here’s pressure put on them if they haven’t got 500 friends, and I think there’s all sorts of online bullying, and I thought, ‘No, this is not really for me.’” This is a purely political stand; this woman would not be a victim of the problems she is raising, and because there is no expectation she would use Facebook, she can easily act on her principles.

Why Focus on Nonuse?

There is a tendency in public discourse and in computing research to view older nonusers as “problematic.”¹⁴ And it is true that by resisting technologies and/or not being able to use them as intended by their designers, older adults impose challenges for realizing the technocentric vision of a fully digital society. But considering nonuse from the perspective of not doing something obscures the fact that nonuse is “active, meaningful, motivated, considered, structured, specific, nuanced, directed, and productive.”¹⁴ Understanding what alternative meaning is conveyed when individuals selectively choose nonuse enables reflection on the meaning(s) implicated or embodied by the technologies the nonusers are rejecting.

Likewise, when older adults say, “Maybe I’m just in a generation where I’d rather go into a bank and speak to someone face-to-face,” they may simply be playing into the common view that they are creatures of habit with nothing better to do as a cover for their prevailing sense of social responsibility. Playing the “age card” to justify rejection of technology is one way older adults take a stand while minimizing the risk of doing so.

It is also often the case that older adults may simply prefer so-called traditional forms of communication, face-to-face, allowing them to ask for the help most of us wish we were entitled to. When it comes to, say, filling in a government form, we have presumably all had the experience this older woman described to us, saying, “I think in my case what happens with the computer is, you’re filling out this bit and this bit and this bit, and sometimes you get so *confused* as to what they’re really asking you.” There is an expectation that younger adults should be able to figure it out themselves, and most will persevere as expected; whereas this woman was empowered by the stereotype to reject the unreasonable demands being placed on her on the basis that she was “too old.” She preferred to walk into her local council office and demand someone answer her question.

We stress that even if future generations of older adults are more digitally adept than today’s older adults, continual technological change alone means they will almost certainly remain less adept than their younger contemporaries. The older adults we talked to often spoke with awe (and an occasional hint of jealousy) about how easily their children, and especially their grandchildren, use technology. There are clear physical and cognitive bases for these observed differences,^{5,6,9} but there are social ones as well, namely that children and younger adults benefit from informal training by their peers and further experiential bases (such as the fact that many of the technologies older adults are most familiar with are becoming “old-fashioned”). It would make sense if older people’s reaction to these observations is to not even try to “compete,” as it were, by working to be as proficient



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with technology as younger people. This excuse of being “too old” will thus continue to be a professed barrier to adoption for a certain segment of the older-adult population. But when and why older adults choose to play the age card may provide clues as to what issues they are protesting, and thus what social side-effects of digital technologies must still be addressed.

Factors Influencing Technology Adoption

Following decades of research focused on getting older adults to adopt technology, there has not been enough progress to ensure older adults are sufficiently adept for navigating a society in which critical services are increasingly “online only.” We suggest this is because the usability and accessibility of these tools, despite being the focus of most research, are not the most salient barriers to adoption. As Zajicek said,²⁰ when there is something they want to do, nothing will get in the way of older adults using technology. This means the more appropriate questions are those that seek to understand what may be underlying older adults’ resistance to developing digital proficiency. While there are definitely cases in which physical and cognitive factors could limit some older adults’ ability to use technology, we maintain there are at least three important factors not often addressed in the literature:

Responsibility. Older adults are uncomfortable with having to take on responsibility for tasks previously handled by trained professionals, particularly when they lack time needed to train themselves sufficiently to perform them with confidence and when genuine risks are associated with using digital technologies improperly;

Values. Older adults make deliberate decisions to not use technologies when they perceive the technology as replacing or eroding something of value to them; and

Cultural expectations. Older adults are one remaining demographic for whom opting out of technology use fits with cultural expectations and thus seems acceptable, despite being increasingly limiting in digital society.

To the extent these factors play a

role in demotivating digital uptake, getting older adults more productively online will require a comprehensive approach that attends to the real-world social and economic consequences of service digitization, explores strategies for de-risking digital technologies, and deeply considers the desirability of the digital world we are asking older adults to inhabit. In order to develop technologies that older adults are able to use, attending to accessibility requirements for those experiencing age-related physical and cognitive decline is a must. But this is clearly not enough. Part of what we have identified is the importance of older adults' perception of the usefulness of technology as a motivator for adoption; but beyond that, we have also found the contextual milieu within which the technology exists must also be understood and addressed. Attending to the concerns central to older adults' resistance to digital technologies should thus not be seen as a matter of accessibility or inclusiveness; we would all be beneficiaries of a more considered approach to digital development that seriously considers how we are able to coexist with technology.

Conclusion

The older adults we interviewed offered a valuable perspective; for most of their lives they functioned just fine without the digital devices and services younger generations take for granted, and they have experienced firsthand the changes digitization has brought. The concerns they raised about digital technologies are valid, and their applicability to younger generations is greatly underappreciated, not least because younger generations will themselves age. Perceptions of greater technical vulnerability that come with aging, and reduced time and energy for maintaining technological proficiency, will likely ensure perception of risk remains a relevant barrier to adoption of new technologies by future generations, even if the particular technologies thought to be risky might change over time. Likewise, while current technologies (such as online banking) could become so essential to daily living as to be universally adopted, universal adoption will

only contribute to future resistance to change when new technologies arrive. And finally, while the specific changes older adults are protesting today may not be a cause for concern for future generations, technological innovation will continue to have wide-reaching societal consequences that may provoke protest among future older adults who resist the loss of whatever it is they value. For such reasons, not only are older adults likely to remain behind the curve in terms of adoption for generations to come and require some degree of accommodation for their relative lack of proficiency, their instances of and justifications for non-use will help draw attention to the trade-offs being made in developing new technologies.

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